FROM THE

EDITOR-IN-CHIEF

Have you ever wondered what happens to your mobile phone after you discard it? The sheer numbers involved make it an urgent question. It is estimated that India has over one billion devices in active use out there. The average life cycle of phones is about three to four years, which means that somewhere there is a planet-sized dump welling up. These devices contain precious metals, rare earth elements and plastic—and extraction presents both an opportunity and an environmental hazard. If one takes the average weight of a phone as 200 grams, that's already 200,000 tonnes of e-waste being generated in the next two to four years. Add to that the production of over 300 million mobile phones every year. E-waste is going to grow exponentially, so will the challenge in its safe disposal.

E-waste has a broader scope than just phones, of course. All three categories of electronic and electrical products—white (refrigerators, washing machines, air-conditioners), grey (desktop computers, laptops, cellphones, printers) and brown (television sets, cameras and recorders)—generate e-waste. The composition of these products goes roughly like this: one half is iron and steel, 21 per cent plastic and 13 per cent copper, aluminium and precious metals. In addition, there are hazardous substances such as mercury, lithium, lead, zinc and cadmium, which, if not handled carefully, can result in severe disease. India is today the world's third-largest producer of e-waste—at 3.2 million tonnes, it trails only China (10 million tonnes) and the USA (6.9 million tonnes). It is estimated that by 2030, India will generate 14 million tonnes of e-waste—quadrupled in just eight years.



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However, modern societies have developed the concept of a circular economy, dedicated to reusing, recycling and regeneration. From what was seen as an environmentalist fancy when coined in the 1980s, the creeping alarm brought about by the climate crisis has made it part of the core lexicon of the industrial world. The NITI Aayog even has a committee on the circular economy. The concerted efforts to regulate e-waste show the seriousness of intent at the government level. After a basic policy framework in 2016, a draft set of new waste rules was circulated this May, proposing a revamp of e-waste management, which widened the range of products from 21 to 95 items.

But practice is often trickier than theory. When it comes to e-waste, the problems are even trickier. There are hazardous substances involved. "You can't have e-waste without mercury and lead. It's there in every device," says Asif Pasha, a recycler in Bengaluru. Capacitors and transformers still contain highly carcinogenic compounds called polychlorinated biphenyls—a ban will come into effect only in 2025. Unregulated disposal means these toxic materials contaminate water, soil and air and find their way into our food. Plus, if not handled properly, they *create* and release deadly new pollutants: over 1,000 harmful substances have been identified as being either components of e-waste or as byproducts of informal ways of e-waste recycling. Beyond pollutants, the challenge extends to a whole bouquet of exotic rare earth elements like yttrium, lanthanum, terbium, neodymium, gadolinium and praseodymium present in every smartphone that are extremely difficult to extract economically.

Now, yttrium and lanthanum are hardly the sort of thing the average kabadiwala will be clued into. They are after the things of obvious value. Gold is the most sought-after. In fact, there happens to be a hundred times more gold in a tonne of electronic waste than in a tonne of gold ore—count up to 280 grams, worth Rs 16 lakh. The ratio for silver is 10 times higher. There's even platinum. That's why the name coined by Japanese mineralogist Hideo Nanjyo—urban mining—has caught on globally. But in the informal sector, urban mining is a dismally crude process. One of the usual methods of extraction is to merely dip electronic equipment into an acid bath. That's extremely harmful—both for the workers who risk chronic organ damage, cancer and neurological complications and for the rest of us down the line.

In this week's cover story, 'E-waste: Mounting Threat', Associate Editor Ajay Sukumaran surveys this grim landscape. The neighbourhood of Mustafabad, a hub for e-waste recycling in north Delhi, is an apt visual metaphor for where we are—a maze of bylanes enveloped with hazardous litter and toxic plastic fumes. Is there a way out? Yes, a new kind of sophisticated factory that can extract the last ounce of material from devices. This sector is only in a nascent stage of evolution. At last count, there were 472 units across India; at present, they manage to process barely 15 per cent of the total e-waste we generate. The rest goes through the kabadiwala route to the sundry Mustafabads out there.

The e-waste management policy announced by the government in 2016 was a significant step forward, making it incumbent on manufacturers to bear the responsibility of managing the waste their products create—or Extended Producer Responsibility (EPR). The collection is key here, and annual targets set for manufacturers have been rising—in the four years leading up to 2021-22, they went up from 80,991 tonnes to 153,889 tonnes just for 13 big brands. This catalysed a process of greater formalisation in recycling, with several Producer Responsibility Organisations (PROs) setting up collection centres—becoming a major link in routing e-waste from informal collectors to proper recycling units. This budding ecosystem faces a fresh challenge now. The new rules proposed in May, besides their positives, also potentially dislodge the PROs from their key node in the chain.

India processed an estimated 1.2 million tonnes of e-waste last year. Overall, there is progress, but we have a long way to go, as manufacturers keep producing new models and the e-waste keeps mounting.

(Aroon Purie)